A Three-Parameter Linear Program for the Equation $x + y = mz$

1 Motivation

The one-parameter and two-parameter linear programs of `latex/3_lp_investigation_notes.tex` and `latex/4_two_parameter_model.tex` plateau well above the best published upper bounds for the density of m-sum-free sets. Concretely, in all observed experiments the one-parameter LP never drops below $1/2$, and the two-parameter LP stalls at $8/15$ for $m = 3$ (see `latex/5_experiments_and_bounds.tex`).

Adding a *third* parameter to the witness model enlarges the affine symmetry group from $\mathrm{GL}_2(\mathbb{F}_p) \ltimes \mathbb{F}_p^2$ to $\mathrm{GL}_3(\mathbb{F}_p) \ltimes \mathbb{F}_p^3$ and produces many more useful forbidden identities. As we show below, already very small three-parameter witness families push the LP relaxation bound strictly below $1/2$.

2 The witness forms

Fix a finite family of affine forms in three parameters,

$$L_r(u, v, w) = a_r u + b_r v + c_r w + d_r, \qquad 1 \le r \le n.$$

For each sign pattern $\varepsilon \in \{\pm 1\}^n$ define the atomic set

$$S_\varepsilon = \bigcap_{r=1}^{n} \big(L_r^{-1}(A)\big)^{\varepsilon_r} \subseteq \mathbb{F}_p^3,$$

with $X^{+1} = X$ and $X^{-1} = \mathbb{F}_p^3 \setminus X$. Its (normalized) density $a(\varepsilon) = |S_\varepsilon|/p^3$ is the LP variable.

As usual the atoms form a probability distribution and marginalize to the density $\delta(A)$ along any nonconstant form:

$$\sum_\varepsilon a(\varepsilon) = 1, \qquad a(\varepsilon) \ge 0, \qquad \delta(A) = \sum_{\varepsilon:\, \varepsilon_{r_0} = +1} a(\varepsilon)$$

for any r_0 with $(a_{r_0}, b_{r_0}, c_{r_0}) \neq 0$.

3 Forbidden triples

Proposition 1. *Assume that* $(x, y, z) \in A^3 \Rightarrow x + y \neq mz$. *If, for a triple* (i, j, k),

$$L_i(u, v, w) + L_j(u, v, w) - mL_k(u, v, w) \equiv 0$$

as an identity in $\mathbb{F}_p^3$, *then every atom with* $\varepsilon_i = \varepsilon_j = \varepsilon_k = +1$ *is empty.*

Coefficientwise, the hypothesis is

$$a_i + a_j = ma_k, \quad b_i + b_j = mb_k, \quad c_i + c_j = mc_k, \quad d_i + d_j = md_k.$$

In the all-zero-shift setting $d_r = 0$ this reduces to three linear equations on the direction vectors $(a_r, b_r, c_r) \in \mathbb{Q}^3$.

4 Affine symmetry on $\mathbb{F}_p^3$

Any affine automorphism $T(u, v, w) = M \cdot (u, v, w)^\mathsf{T} + t$ with $M \in \mathrm{GL}_3(\mathbb{F}_p)$ and $t \in \mathbb{F}_p^3$ preserves the uniform distribution on $\mathbb{F}_p^3$, hence

$$\delta\Big(\bigcap_{r \in S} L_r^{-1}(A)\Big) = \delta\Big(\bigcap_{r \in S'} L_{r'}^{-1}(A)\Big)$$

whenever two subfamilies $\{L_r\}_{r \in S}$ and $\{L_{r'}\}_{r' \in S'}$ are related by an affine reparameterization of (u, v, w).

Rank-based normalization

Given a subset of forms, the coefficient vectors (a_r, b_r, c_r) span a linear subspace of $\mathbb{Q}^3$ of some rank $r \in \{0, 1, 2, 3\}$. Under the full action of $\mathrm{GL}_3(\mathbb{F}_p) \ltimes \mathbb{F}_p^3$, we may always rotate coordinates so that this subspace is $\mathbb{Q}^r \times \{0\}^{3-r}$. After this, the affine model collapses to an r-parameter affine model, and the canonical shape can be computed by the existing r-parameter logic:

- Rank 0 (all forms constant): shape is the multiset of constant terms.
- Rank 1: project onto the 1-dimensional span and apply the one-parameter canonical shape (`affine_form_lp.canonical_subset_shape`).
- Rank 2: choose any two independent forms as a basis, expand every other form in that basis; apply the two-parameter canonical shape (`two_parameter_lp.canonical_subset_shape`).

- Rank 3: try every triple of independent forms as a GL_3 basis, normalize to (u, v, w), and take the lexicographically smallest tuple.

Tagging the canonical shape with its rank ensures that two subsets of different ranks are never identified.

5 The resulting LP

Given a family $\mathcal{L} = \{L_1, \ldots, L_n\}$, the three-parameter LP has

- variables $a(\varepsilon) \geq 0$ for $\varepsilon \in \{\pm 1\}^n$;
- the mass constraint $\sum_\varepsilon a(\varepsilon) = 1$;
- for every identity $L_i + L_j - mL_k \equiv 0$ and every ε with $\varepsilon_i = \varepsilon_j = \varepsilon_k = +1$, the constraint $a(\varepsilon) = 0$;
- for every pair of subsets $S, S' \subseteq \{1, \ldots, n\}$ with the same canonical shape, the equation
$$\sum_{\varepsilon:\, \varepsilon|_S = +1} a(\varepsilon) = \sum_{\varepsilon:\, \varepsilon|_{S'} = +1} a(\varepsilon);$$
- the objective $\max \sum_{\varepsilon:\, \varepsilon_{r_0} = +1} a(\varepsilon)$.

6 Implementation

The module `three_parameter_lp.py` implements this model end-to-end:

- `linear_form_3d(a,b,c,d,label)` stores a form $au + bv + cw + d$ with rational coefficients.
- `canonical_subset_shape(forms)` computes the rank-tagged canonical shape of a subset. Rank 1 and rank 2 subsets are reduced to the one-parameter and two-parameter canonical shapes respectively.
- `form_relation_holds(L_i, L_j, L_k, m)` checks whether the identity $L_i + L_j - mL_k \equiv 0$ holds.
- `build_three_parameter_lp(m, forms)` assembles the LP above and solves it via SciPy's HiGHS backend.
- `evaluate_three_parameter_bound(m, forms)` returns the LP upper bound, or `None` if the solver fails.

7 Small illustrative families

The following three families are defined in `experiments_three_parameter.py`.

`basic_pair_family`(m). Six forms

$$u, \quad v, \quad w, \quad \frac{u+v}{m}, \quad \frac{u+w}{m}, \quad \frac{v+w}{m}.$$

Three forbidden triples: $u + v = m \cdot (u + v)/m$, and cyclic rotations. The LP optimum is

$$\boxed{\delta(A) \leq \frac{16}{33} \approx 0.4848}$$

for every $m \geq 3$ and is *independent* of m.

`signed_pair_family`(m). Adds the three "minus" variants

$$\frac{u-v}{m}, \quad \frac{u-w}{m}, \quad \frac{v-w}{m}.$$

The same three forbidden triples hold (the new forms do not participate in any identity $L_i + L_j = mL_k$ inside this family), but the symmetry content grows substantially. The LP optimum is

$$\boxed{\delta(A) \leq \frac{48}{103} \approx 0.4660}$$

uniformly in $m \geq 3$.

`symmetric_12_family`(m). Further adds the three "$(2a-b-c)/m$" forms:

$$\frac{2u-v-w}{m}, \quad \frac{-u+2v-w}{m}, \quad \frac{-u-v+2w}{m}.$$

Still only three forbidden triples, but the added symmetry drops the LP optimum to approximately

$$\delta(A) \leq 0.4398 \ (m = 3), \qquad 0.4386 \ (m = 4).$$

8 Why the bound is uniform in m for the smaller families

The six and nine-form families use only forms of the form $(\alpha u+\beta v+\gamma w)/m$ with $(\alpha,\beta,\gamma)\in\mathbb{Z}^3$ and $d_r=0$. The change of coordinates $(u,v,w)\mapsto(mu,mv,mw)$ (invertible for $\gcd(m,p)=1$) sends this family to an isomorphic family of integer forms for every m. It maps forbidden identities to forbidden identities (because $L_i+L_j=mL_k$ scales consistently), and the ranks and canonical shapes are invariants of the affine equivalence class. Consequently the LP is the same linear program up to a relabeling of variables, so the optimum does not depend on m for $m\geq 3$.

9 Remarks and caveats

Remark 1. *These are* LP relaxation *bounds for joint atomic laws compatible with affine symmetry and the empty-atom constraints. Turning them into theorems about m-sum-free sets $A\subset\mathbb{F}_p$ requires the usual step of realizing the LP from real set statistics, as in [1].*

Remark 2. *With n forms the LP has at most 2^n atomic variables; in practice the twelve-form family already produces* 2 880 *variables after the forbidden atoms are excluded and about* 1 700 *shape equations, and SciPy's HiGHS solves it in a few minutes.*

Remark 3. *The rank-based canonical shape is essential: the earlier three-parameter LP stored residual w-coefficients for rank-2 subsets and therefore treated "tilted" and "axis-aligned" 2-planes as different orbits, losing some symmetry equations. After projecting rank-1 and rank-2 subsets to their natural low-dimensional models, the LP correctly identifies those subsets and the bounds tighten from* $21/41$, $11/23$ *and* 0.449 *down to the values listed above.*

References

[1] V. Lev, M. Matolcsi, P. P. Pach, D. Varga, *On the density of Kravitz sets*, `https://arxiv.org/pdf/2510.16522`.